

Effect of diet on excretion of estrogens in pre- and postmenopausal women.

Fecal, urinary, and plasma estrogens and plasma androgens were studied in healthy pre- and postmenopausal vegetarian and omnivorous women. Dietary histories of the subjects revealed that omnivores consumed a higher percentage of total protein and fat from animal sources. The total 72-hr fecal excretion as measured by dry weight was higher for vegetarians. Preliminary results indicate that vegetarian women excrete 2 to 3 times more estrogens in feces than do omnivores and that omnivores have about 50% higher mean plasma level of unconjugated estrone and estradiol than vegetarians. Estriol-3-glucuronide, a compound that is formed upon reabsorption of free estriol from the intestine, is found in lower concentrations in the urine of vegetarians. These data suggest that in vegetarians a greater amount of the biliary estrogens escape reabsorption and are excreted with the feces. The differences in estrogen metabolism may explain the lower incidence of breast cancer in vegetarian women.